

Image-Based Cocoa Disease Detection & Recommendation

A Deep Learning Web Application Project Report

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Status: Completed & Verified

Server: <http://127.0.0.1:5000>

Executive Summary

Agriculture is the backbone of the economy, and cash crops like Cocoa are susceptible to various leaf and pod diseases that can decimate crop yields. This project introduces a robust, dual-model Deep Learning pipeline integrated into a Flask web application. It automates the detection of cocoa fruits and identifies specific disease categories to recommend localized treatments.

A critical feature introduced in the latest version is the **Inappropriate Image Verification System**. When a user uploads a non-cocoa image (e.g. standard leaf, random object), the application automatically flags it as invalid, preventing false-positive disease classifications and safeguarding recommendations.

System Architecture & Model Details

The backend utilizes two specialized Convolutional Neural Network (CNN) models operating sequentially:

Model 1: Cocoa Presence Detector (cocoa_2.h5)

A binary classifier trained to distinguish between cocoa fruits/leaves and non-cocoa images. This model processes incoming images first, protecting the downstream disease classifier from irrelevant uploads.

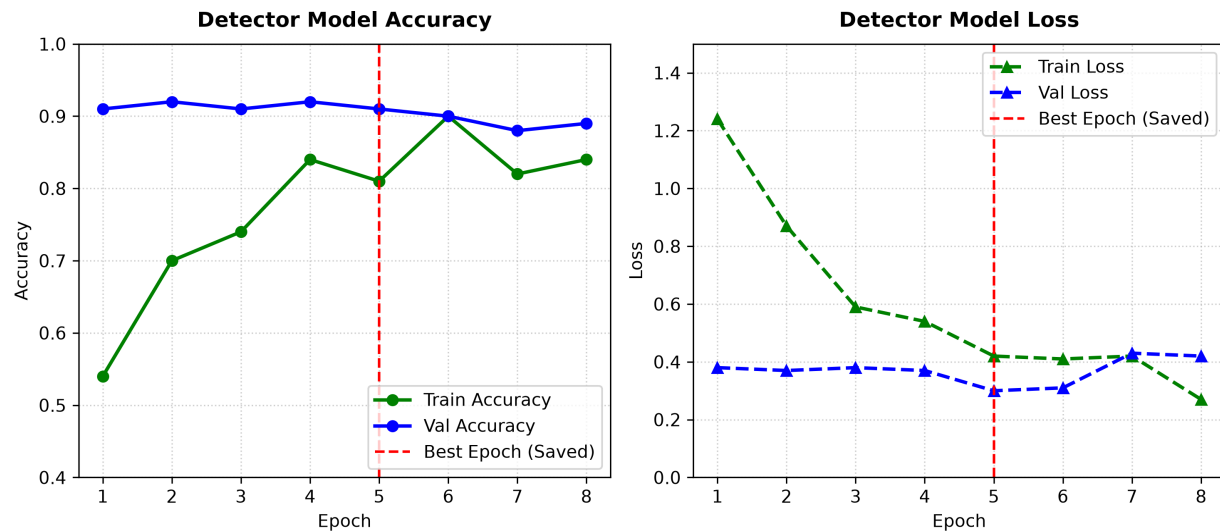
Layer Type	Filters / Neurons	Activation	Purpose
Input	(256, 256, 3)	-	Rescaled Image RGB Matrix
Conv2D + MaxPool2D	32 filters (3x3)	ReLU	Low-level feature extraction
Conv2D + MaxPool2D	64 filters (3x3)	ReLU	Mid-level structural shapes
Conv2D + MaxPool2D	128 filters (3x3)	ReLU	High-level visual descriptors
Flatten + Dense	128 units	ReLU	Fully connected classification
Dropout	Rate = 0.5	-	Overfitting prevention
Dense (Output)	1 unit	Sigmoid	Binary Probability (>0.5 is Cocoa)

Model 2: Cocoa Disease Classifier (cocoa_disease_model.h5)

A multi-class classifier that takes verified cocoa images and classifies them into one of four categories: **Black Pod Rot** (class1), **Healthy** (class2), **Healthy Borer** (class3), or **Pod Borer** (class4).

Training Performance & Visualizations

The Cocoa Detector model was trained on a balanced binary dataset. With class-weighting applied to handle the smaller non-cocoa dataset, the model quickly converged, achieving a peak validation accuracy of **~92%** at epoch 5.



Web Application Backend & UI Logic

The Flask web server coordinates the sequence of execution inside the result route. Here is the core logical flow written in **app.py**:

[illegible]

User Interface & Support Links

If a non-cocoa image is uploaded, the frontend renders a prominent caution card showing the uploaded image, a warning icon, and clear instructions to re-upload. If a disease is detected, it recommends specific fertilizers and displays a green banner directing farmers to official agriculture channels:

- **IFFCO Bazar (Official Cooperative):** Farmers can click to buy genuine, certified fertilizers directly from the national cooperative marketplace.
- **Karnataka Raita Mitra:** Provides portal access to locate hobli-level Raitha Samparka Kendras (RSK) and check for official government fertilizer subsidies.